Next.js Project Learning Topics

A focused study guide for the graduation project presentation. Covers the Next.js, React, TypeScript, validation, API, styling, and library topics used in the admin dashboard.

1. Core Next.js Topics

App Router
The app folder structure, page.tsx files, layout.tsx files, nested layouts, and route-driven UI.

Dynamic routes
Folder names such as [locale] and how URLs like /en/dashboard and /ar/dashboard are handled.

Route groups
Parentheses folders such as (auth), (dashboard), and (standalone). They organize code without changing the URL path.

Client components
The use client directive and why React hooks like useState and useEffect require client components.

Server components
Default component behavior in Next.js when use client is not added.

Navigation
Link, useRouter, usePathname, and useParams from Next.js navigation.

Middleware
How middleware can run before a route loads. In this project it supports locale routing.

Environment variables
NEXT_PUBLIC variables, API base URLs, and how frontend configuration is passed safely.

Public assets
How images, SVGs, and static files from public are referenced in pages and components.

Next fonts
Using next/font/google to load optimized fonts.

2. React APIs Used Inside The Next.js App

Components - React
UI is split into reusable pieces with props and JSX.

State - useState
Stores values that change on the client, such as form fields, filters, and selected rows.

Effects - useEffect
Runs side effects such as loading API data after a component renders.

Memoized values - useMemo
Avoids recalculating derived values unless inputs change.

Stable callbacks - useCallback
Keeps function references stable, useful for effects and child components.

DOM references - useRef
Stores references to DOM nodes, timers, or values that should not trigger re-rendering.

Context - createContext / useContext
Shares state across related screens, such as multi-step signup data.

Custom hooks - Project hooks
Reusable logic functions like useBookings, usePayments, and useDashboard.

3. TypeScript Topics

Types and interfaces
Describe the shape of data like Booking, Payment, Report, and API responses.

Generics
Reusable typed components and functions, such as a table that can display different row types.

Union types
Limited sets of allowed values, such as payment methods, report periods, and export formats.

Optional fields
Handling values that may be missing, null, or undefined.

Path aliases
Imports like @/application/hooks/useBookings instead of long relative paths.

4. Validation And Forms

Schema validation - Zod
Define rules for email, password, owner information, business information, bank details, and MFA.

Controlled forms - React state
Inputs read from state and update state on every change.

Submit handling - onSubmit / async functions
Validate data, call APIs, handle loading, then navigate or show errors.

Error messages - Conditional rendering
Show field errors and form errors only when validation or API calls fail.

Multi-step forms - React Context + router
Keep signup data while moving between onboarding screens.

5. API, Authentication, And Backend Communication

REST APIs
GET, POST, PUT, PATCH, DELETE and how frontend actions map to backend endpoints.

Fetch API
How browser code sends requests, attaches headers, serializes JSON, and reads responses.

API client pattern
A shared helper builds URLs, sends requests, handles JSON, errors, auth headers, and token refresh.

JWT / token auth
Access tokens, refresh tokens, bearer authorization headers, and why tokens expire.

Local storage
Where the browser stores the auth session for client-side access.

Cookies
How an auth token can also be written as a cookie for route/session handling.

OpenAPI
A machine-readable backend contract that documents endpoints, request bodies, and responses.

6. Styling And UI Libraries

Scoped styling - CSS Modules
Component-specific CSS files that avoid global class conflicts.

Global styles - globals.css
CSS variables, shared classes, fonts, and base styling.

Utility styling - Tailwind CSS
Utility-first CSS approach, even if most styling is done with modules.

Icons - React Icons / Lucide React
Reusable icon components for navigation, buttons, status, and actions.

Material UI support - MUI + AppRouterCacheProvider
Integration support for Material UI styles in the Next App Router.

Responsive design - CSS
Flexbox, grid, breakpoints, overflow behavior, and dashboard layouts.

7. Feature-Specific Libraries And Concepts

Internationalization - next-intl
Locale routing, messages, English/Arabic support, and RTL/LTR awareness.

Push notifications - Firebase Cloud Messaging (FCM)
Notification permission, Firebase web messaging, service worker, VAPID key, device token registration, and backend delivery.

Phone input - react-international-phone
Country-aware phone number inputs and formatting.

Charts - SVG + React
How report data becomes plotted points, lines, labels, and tooltips.

Tables - Custom components
Pagination, sorting, filtering, searching, row selection, and bulk actions.

Toasts - Custom notification UI
Temporary success/error messages after user actions.

8. Suggested Learning Priority

1. Next.js App Router, page.tsx, layout.tsx, dynamic routes, and route groups.
2. Client components and the use client directive.
3. React hooks: useState, useEffect, useMemo, useCallback, useRef, useContext.
4. TypeScript types, interfaces, generics, and union types.
5. Zod validation and controlled forms.
6. REST APIs, fetch, API client helpers, and error handling.
7. Authentication tokens, local storage, cookies, and protected route behavior.
8. next-intl localization for English and Arabic routes.
9. CSS Modules, global CSS variables, icons, and responsive dashboard UI.
10. Firebase Cloud Messaging: notification permission, service worker, VAPID key, device tokens, and backend notification flow.
11. Reusable components, custom hooks, and how data flows from page to hook to service to API.

Note: useState is a React hook, not a separate library. Zod, next-intl, React Icons, Lucide React, Firebase, Tailwind CSS, MUI, and react-international-phone are external libraries used alongside Next.js and React.